Fundamentals of Business Economics IS 2211

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Lecture 3
Demand and Supply Together

Learning Outcomes

At the end of the session, you should be able to:

- Explain what is market equilibrium
- Explain what is market disequilibrium
- Find the market equilibrium
- Analyze changes in market equilibrium
- Explain how prices allocate resources

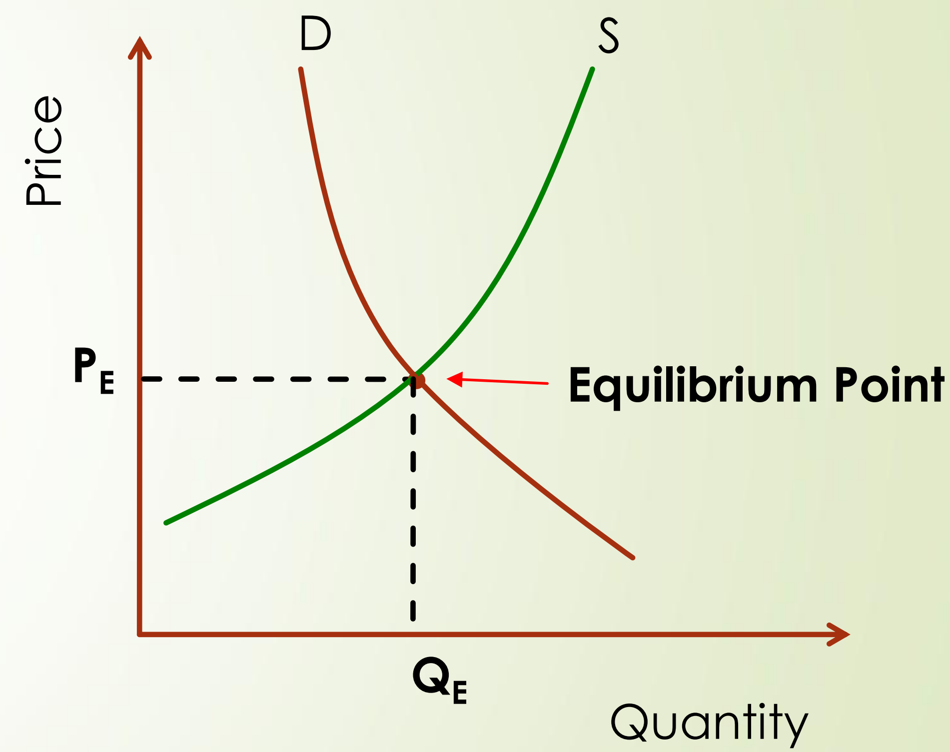
Demand and Supply Together

Having analyzed supply and demand separately, we now combine them to see how they determine the quantity of a good bought and sold in a competitive market and price.

- **Markets function through the interaction of demand and supply**, influencing prices and quantities.
- **Neither demand nor supply alone determines the outcome**, they must be considered together.
- **Prices act as signals**, adjusting based on consumer demand and producer supply.
- **When demand and supply interact, the market moves toward a balance**, setting the stage for equilibrium.

Market Equilibrium – Key Concepts

- **Market Equilibrium**
  - Market equilibrium is the point where demand and supply balance each other, determining the price and quantity of goods exchanged in a market.
  - At this point, the quantity demanded by consumers equals the quantity supplied by producers, meaning there is no persistent shortage or surplus.
- **Equilibrium Price (P*)** → The price at which demand and supply are balanced.
- **Equilibrium Quantity (Q*)** → The quantity exchanged in the market at the equilibrium price.



Market Equilibrium – How it works

- Prices naturally adjust to reach equilibrium: if there is excess supply, prices tend to fall; if there is excess demand, prices tend to rise.
- Once equilibrium is reached, the market is stable unless external factors (such as technological changes, government policies, or consumer preferences) shift demand or supply.
- Understanding equilibrium helps explain how markets function efficiently, ensuring optimal resource allocation and price stability.

Example
Cloud Storage Market:
•**Demand Surge:** As more businesses adopt cloud storage for data management, demand increases. Initially, prices may rise due to high demand, creating a shortage of available storage.

•**Supply Response:** Cloud providers respond by increasing capacity (e.g., adding servers or expanding data centers) to meet demand. As supply increases, prices start to stabilize, moving toward the equilibrium price.

•**Equilibrium Reached:** Over time, the increased supply and stabilized prices result in the **equilibrium price** (P*) and **equilibrium quantity** (Q*) being reached, where the amount of storage consumers want to buy equals the amount cloud providers are willing to sell

Finding Market Equilibrium

- In reality, supply and demand curves are often not perfectly linear.
- They can exhibit various shapes and characteristics, such as being curved, having multiple equilibrium points, or even being discontinuous in certain cases.
- However, despite these complexities, economists often use linear approximations for supply and demand curves for several reasons such as simplicity, generalization, and ease of interpretation, etc.

Finding Market Equilibrium – Examples

Example 1: Cloud Storage Services (Linear)

A cloud storage provider wants to determine the equilibrium price and quantity for its storage plans. The company collects data and finds that:

The demand function for cloud storage is:
Q_d=500-5P where Q_d represents the number of storage plans customers are willing to buy at a given price P(in dollars).

The supply function for cloud storage is:
Q_s=100+3P where Q_s represents the number of storage plans cloud providers are willing to offer at price P

1.Find the equilibrium price (P∗)

2.Find the equilibrium quantity (Q∗) at this price.

3.Interpret the result: What does it mean for the cloud storage market?

Finding Market Equilibrium - Examples

Example 2: Smartphone Market Equilibrium (Non-Liner)

A smartphone company wants to determine the equilibrium price and quantity for its new smartphone model. The company collects data and finds the following:

•**Demand Function**:

•$Q_d = 300 - P^2$ where Q_d is the quantity demanded, and P is the price per smartphone (in dollars).

•**Supply Function**:

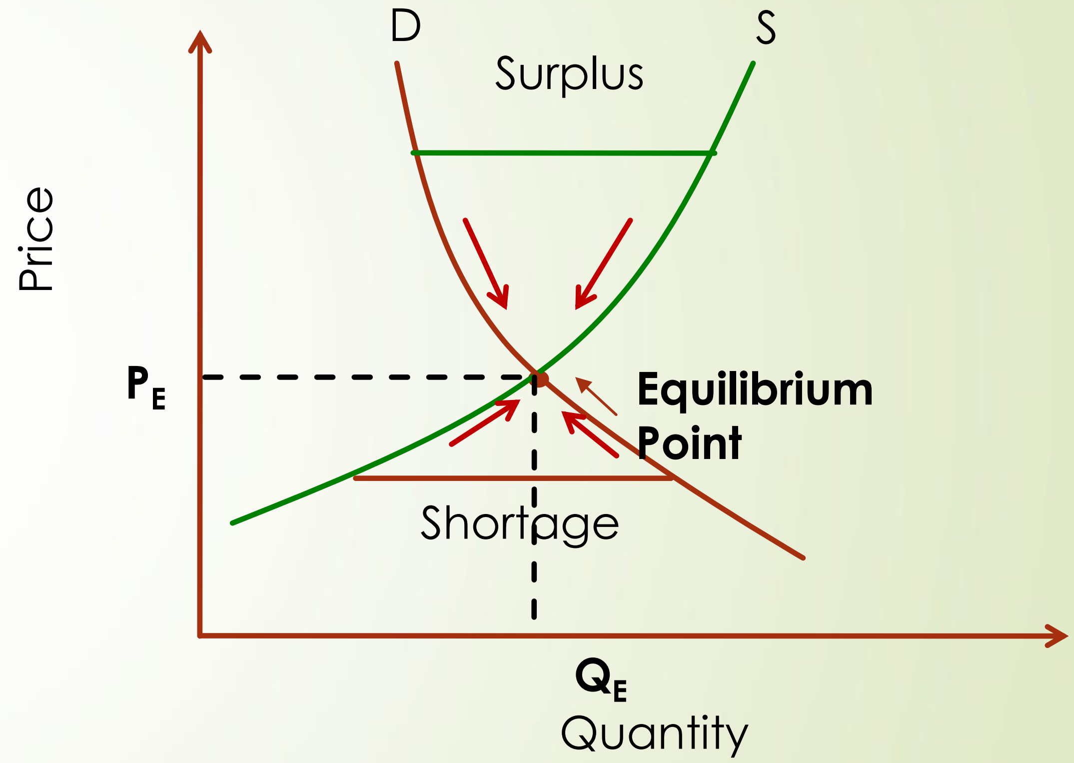
$Q_s = 100 + P^2$ where Q_s is the quantity supplied.

1.Find the **equilibrium price** (P∗).

2.Find the **equilibrium quantity** (Q∗) at this price.

Disequilibrium in a Market

- Disequilibrium occurs when quantity supplied is not equal to the quantity demanded
- When a market is experiencing a disequilibrium, there will be either a shortage or a surplus.
- In both cases, economic forces work to bring the market back to equilibrium, where supply equals demand.

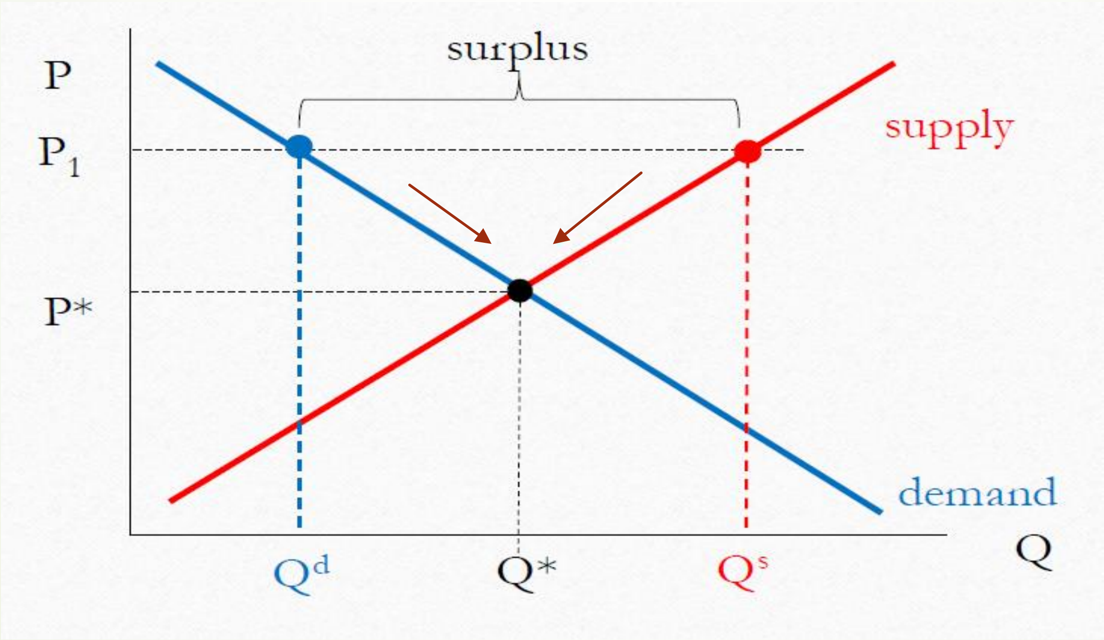


Disequilibrium in a Market - Surplus

A **surplus** occurs when the quantity supplied (Qs) is **greater** than the quantity demanded (Qd) at a given price. This happens when the price is set **above the equilibrium price**.

- **Causes of Surplus:**

- The price is too high, discouraging buyers,
- Overproduction by suppliers,
- A decrease in consumer demand (e.g., due to changes in preferences, income, or availability of substitutes.



To address the surplus, sellers may lower prices to incentivize buyers to purchase more goods.

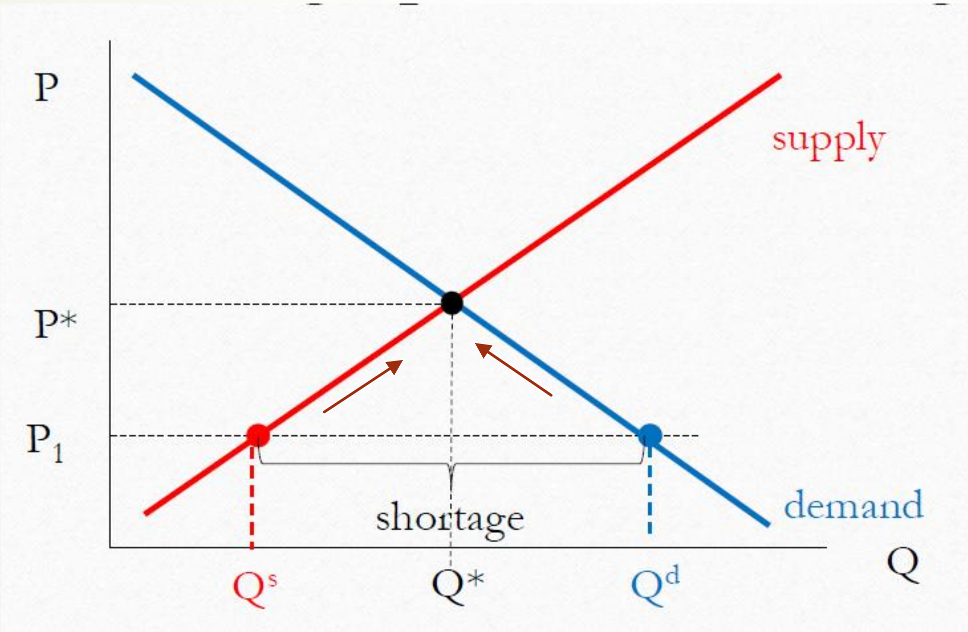
As prices decrease, more buyers are attracted to the market, and fewer sellers find it profitable to produce at lower prices.

This decrease in price continues until it reaches a level where the quantity demanded equals the quantity supplied, thus restoring equilibrium.

Disequilibrium in a Market - Shortage

A **shortage** occurs when the quantity demanded (Qd) is **greater** than the quantity supplied (Qs) at a given price. This happens when the price is set **below the equilibrium price**.

- **Causes of Shortage:**
- The price is too low, making the product more attractive to consumers.
- Limited production or supply chain disruptions.
- A sudden increase in demand (e.g., due to trends, new product launches, or external factors).

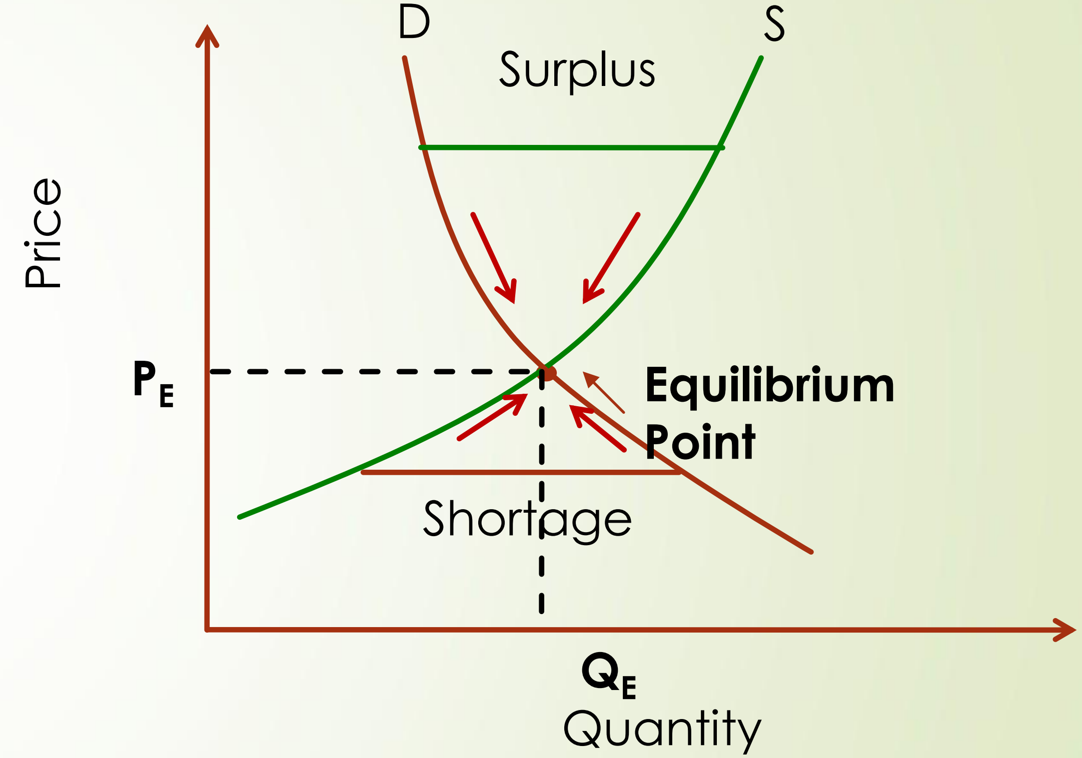


To address the shortage, sellers may raise prices due to increased demand or scarcity of goods. As prices rise, some buyers may reduce their quantity demanded, while more sellers find it profitable to produce at higher prices, increasing the quantity supplied.

This increase in price continues until it reaches a level where the quantity demanded equals the quantity supplied, thus restoring equilibrium.

Disequilibrium in a Market (Cont'd.)

- The activities of many buyers and sellers push the market price towards the equilibrium price.
- Once the market reaches its equilibrium, all buyers and sellers are satisfied and there is no upward or downward pressure on the price.
- How quickly equilibrium is reached varies from market to market, depending on how quickly the prices adjust.
- In most free markets, surpluses and shortages are only temporary, because prices eventually move towards their equilibrium.



Example

A technology company produces and sells wireless routers. The company has observed the following data for the market:

•The **demand function** for wireless routers is: Q_d=400-4P
where Q_d is the quantity demanded at a price *P* (in dollars).

•The **supply function** for wireless routers is: Q_s=50+3P
where Q_s is the quantity supplied at a price *P* (in dollars).

Given that the company has set the price of the wireless router at **$70**:

1.Find the quantity demanded and supplied at the price of $70.

2.Determine if there is a shortage or surplus at this price.

3.Calculate the size of the shortage or surplus.

Changes in Equilibrium - Change in Demand

- Imagine that the market starts out in equilibrium and then, there is a change in demand.
- If that change in demand is instantaneous, the market will be out of the old equilibrium and will gravitate toward a new equilibrium.

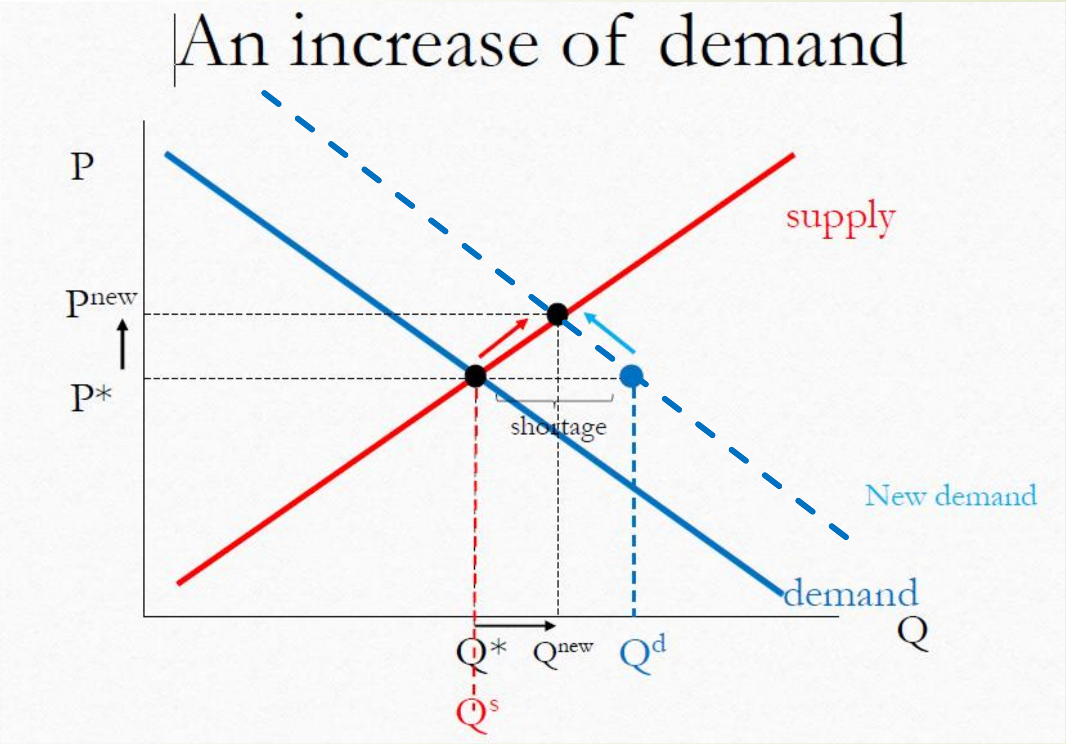
Example :Increase in Demand

Cloud Storage Services

Imagine a cloud storage provider that offers online storage plans. Initially, the market is in equilibrium at a price of $10 per month with 10,000 subscribers.

What happens if demand increases?

- A new trend in remote work and data security awareness increases the number of people needing cloud storage. As a result, more customers want to buy storage plans at the same price. The demand curve shifts rightward.
- At the original price, there is now a shortage (more demand than supply).To restore equilibrium, the price of cloud storage increases, and providers expand their server capacity to meet the new demand.
- A new equilibrium is established at a higher price and a larger quantity of cloud storage plans sold .More cloud storages are sold at a higher price.



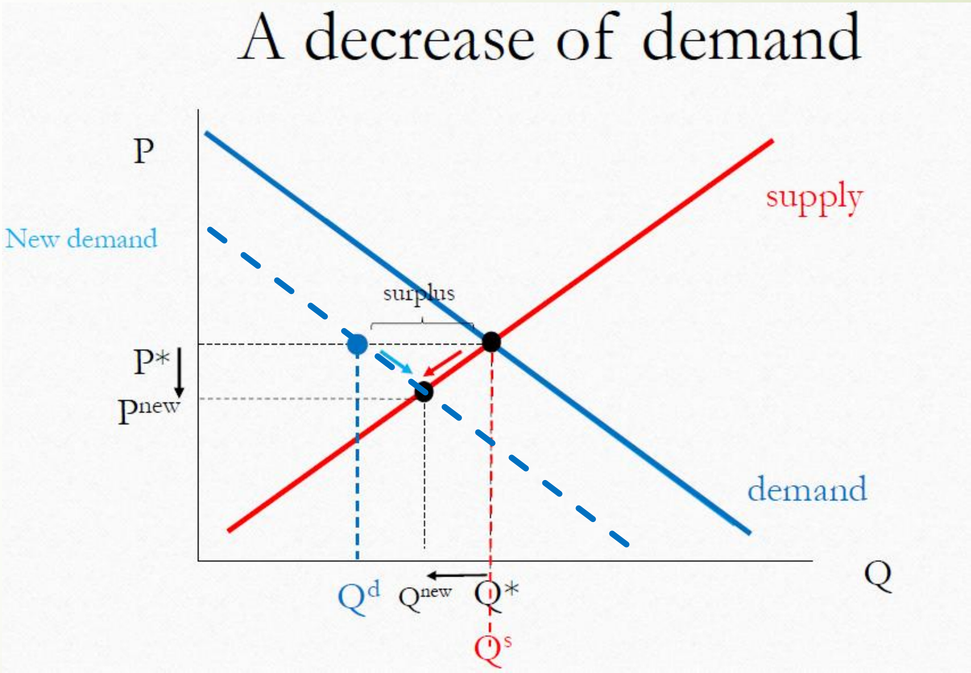
Changes in Equilibrium - Change in Demand

Example : Decrease in Demand

Smartphones with Outdated Technology
Consider a smartphone manufacturer that sells a model for $800 with 50,000 units sold per month in equilibrium.

What happens if demand decreases?

- A newer model with advanced features is released by competitors. Consumers shift to the newer model, and fewer people are willing to buy the older version at the original price.
- The demand curve shifts leftward. At the original price, there is now a surplus (more supply than demand).To restore equilibrium, the company lowers the price to attract more buyers.
- A new equilibrium is established at a lower price and lower quantity sold.



Changes in Equilibrium - Change in Supply

- Imagine that the market starts out in equilibrium and then, there is a change in supply.
- If that change in supply is instantaneous, the market will be out of the old equilibrium and either have a surplus or a shortage (depending what the change in supply was).
- With time, if the market is left on its own, it will gravitate toward a new equilibrium.

❖ Increase of Supply

What can cause an increase of Supply?

- A decrease in the price of an input
- Technological improvement (or extra resources)
- Sellers expect future prices to drop
- A decrease in the price of a substitute in production
- An increase in the number of sellers

Change in Supply – Increase of Supply

Example
Increase in Supply: Cloud Computing Services Become More Efficient

Imagine a **cloud computing provider** that offers virtual machines and cloud storage. Initially, the market is in equilibrium at a price of **$15 per month** with **100,000 customers** subscribing to the service.

What happens if supply increases?

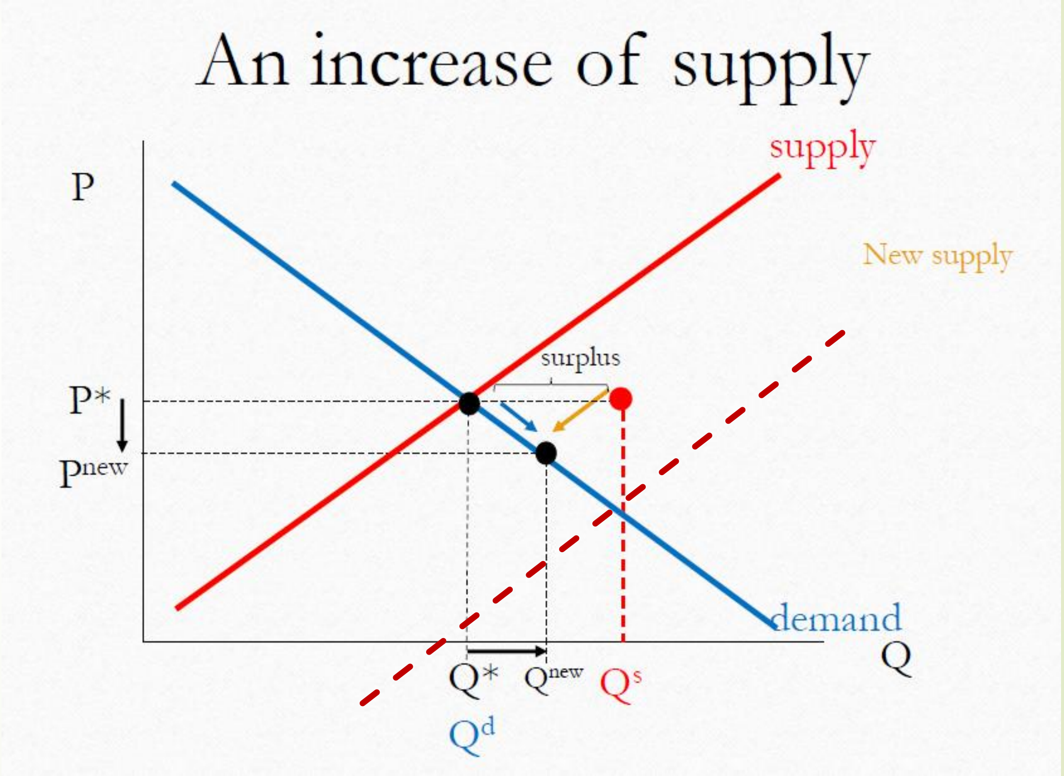
•A **new technology** improves data center efficiency, reducing costs for cloud providers. More companies enter the market, increasing the availability of cloud services.

•The **supply curve shifts rightward** (more supply at each price).

•At the original price, there is now a **surplus** (more supply than demand).

•To sell the excess cloud services, providers **reduce prices**, making the service more affordable.

•A **new equilibrium is established** at a **lower price and higher quantity** of cloud services.



Change in Supply – Decrease of Supply

Example
Decrease in Supply: Semiconductor Shortage in the Laptop Market

Consider a **laptop manufacturer** that sells a model for **$1,200** with **200,000 units sold per month** in equilibrium.

What happens if supply decreases?

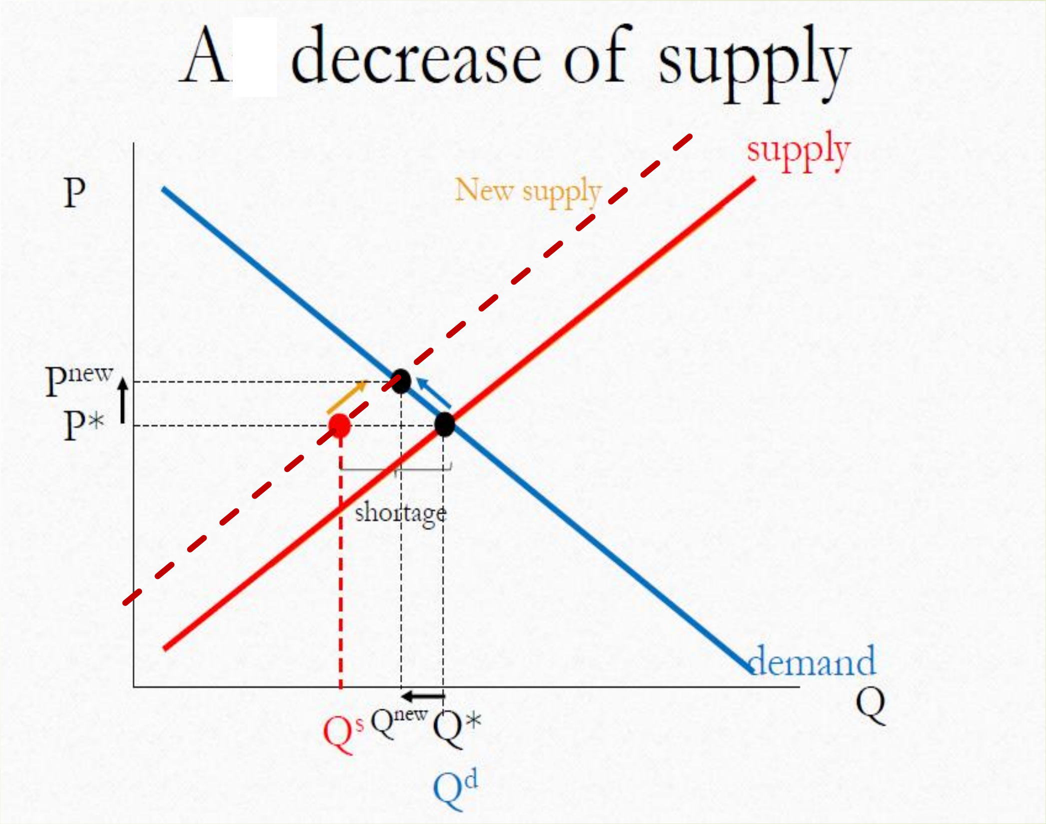
•A **global semiconductor shortage** occurs due to supply chain disruptions, reducing the availability of key components. Laptop manufacturers struggle to produce as many units as before.

•The **supply curve shifts leftward** (less supply at each price).

•At the original price, there is now a **shortage** (more demand than supply).

•As a result, the company **increases prices**, and some customers choose to delay their purchases.

•A **new equilibrium is established** at a **higher price and lower quantity sold**.



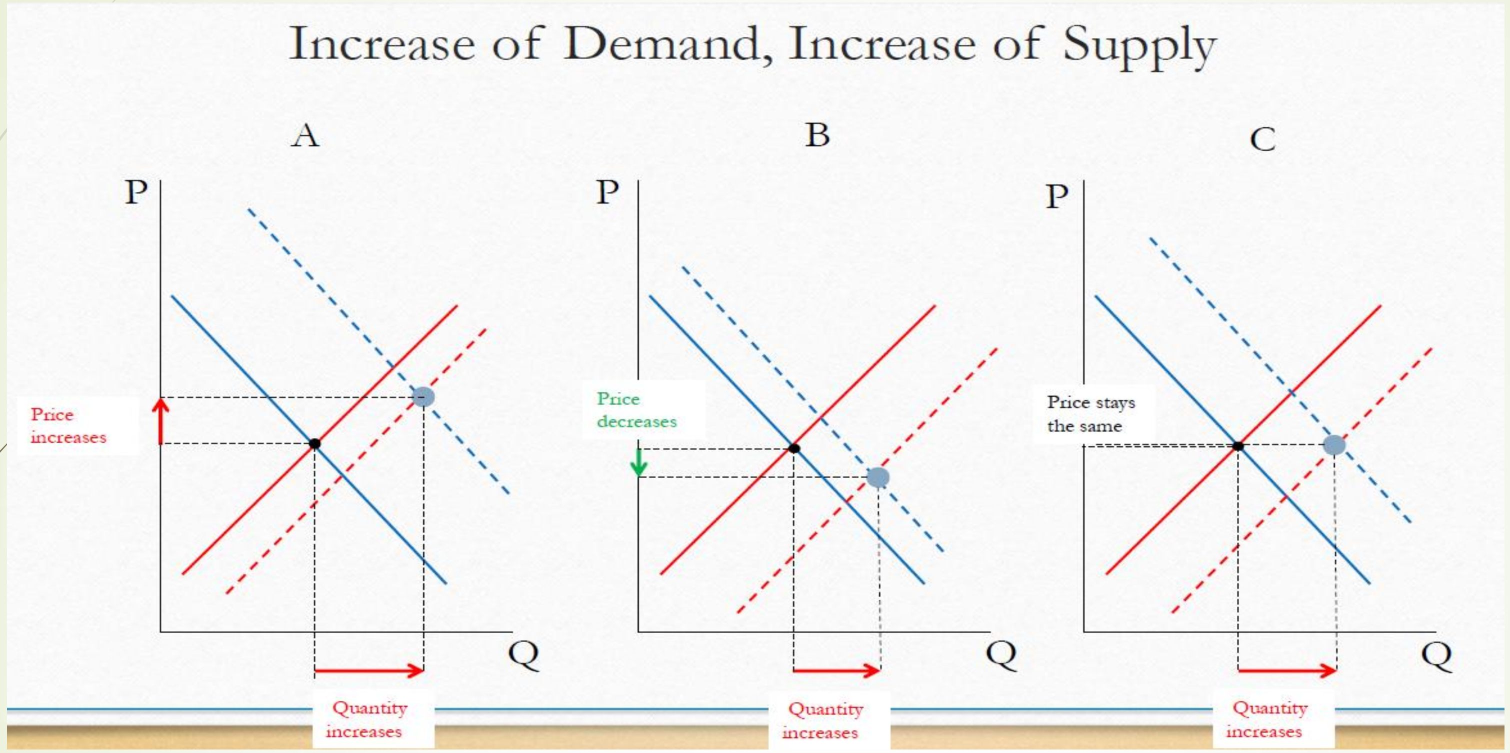
Changes in Equilibrium – Complex Changes

We are analyzing the impact when both the demand and the supply change simultaneously.

1) Demand increases, Supply increases
2) Demand increases, Supply decreases
3) Demand decreases, Supply increases
4) Demand decreases, Supply decreases

Changes in Equilibrium – Complex Changes

1) Demand increases and Supply increases

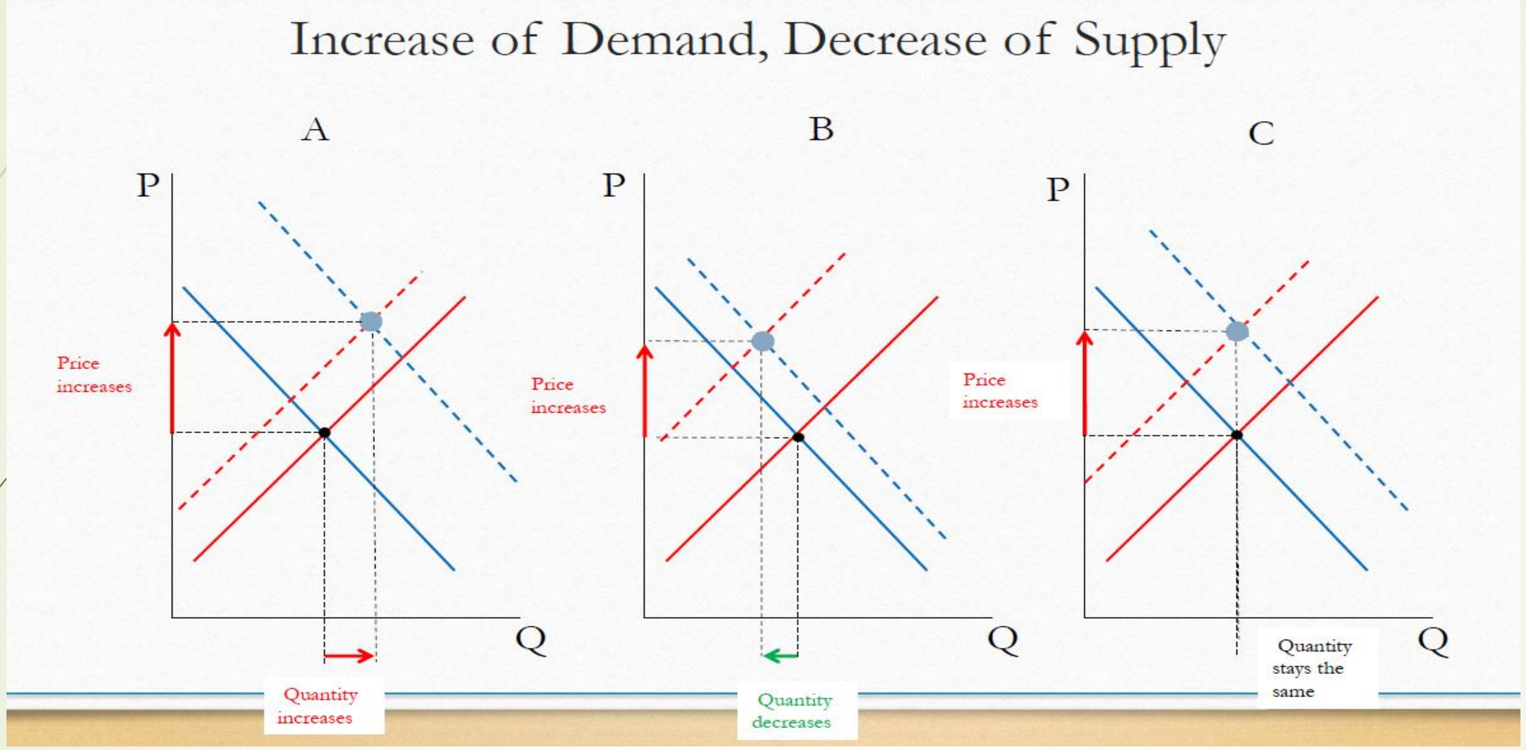


Effect on Price and Quantity When Both Demand & Supply Increase
Quantity Always Increases
•Since both demand and supply increase, **the equilibrium quantity will definitely increase**.
•More goods/services are exchanged in the market.

Price Depends on the Magnitude of the Shifts
•If **demand increases more than supply**, the price **rises**.
•If **supply increases more than demand**, the price **falls**.
•If both increase **equally**, the price may **stay the same**.

Changes in Equilibrium – Complex Changes

2) Demand increases and Supply decreases

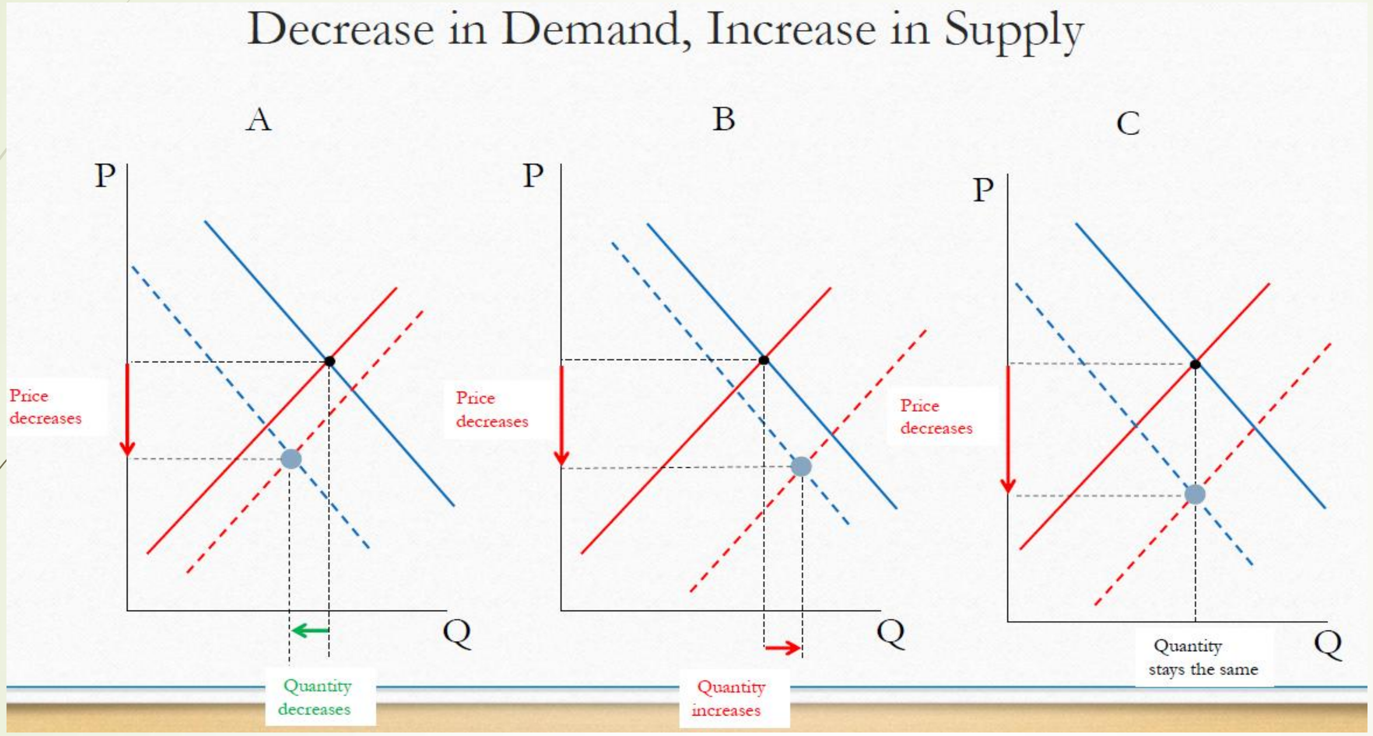


Effect on Price and Quantity When Demand Increases & Supply Decreases
Price Always Increases
•Higher demand puts **upward pressure on price**.
•Lower supply creates **a shortage**, further **raising the price**.

Quantity Depends on the Magnitude of the Shifts
•If **demand increases more than supply decreases**, the quantity **increases.**
•If **supply decreases more than demand increases**, the quantity **decreases**.
•If both increase **equally**, the quantity may **stay the same**.

Changes in Equilibrium – Complex Changes

3) Demand decreases and Supply increases



Price Always Decreases

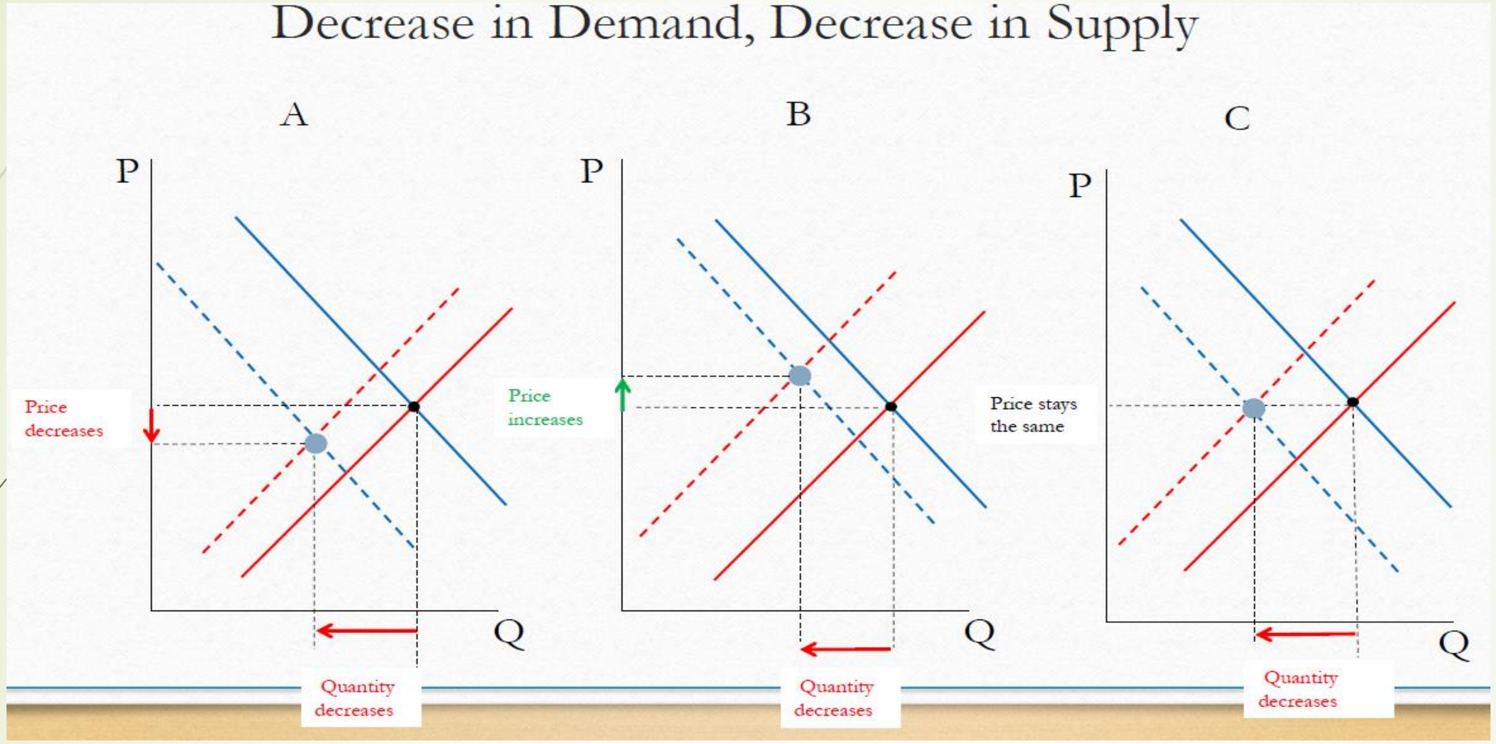
•Lower demand **reduces willingness to pay**.
•Higher supply **creates a surplus**, pushing the price down.

Quantity Depends on the Magnitude of the Shifts

•If **demand decreases more than supply increases**, the quantity **decreases**.
•If **supply increases more than demand decreases**, the quantity **increases**.

Changes in Equilibrium – Complex Changes

4) Demand decreases and Supply decreases



Quantity Always Decreases
•Less demand = **fewer people willing to buy**.
•Less supply = **fewer goods available in the market**.

Price Depends on the Magnitude of the Shifts
•If **demand decreases more than supply**, the price **falls**.
•If **supply decreases more than demand**, the price **rises**.

Changes in Equilibrium (Cont'd.)

Summary

Change	Change in P^*	Change in Q^*
Supply increases ↑ (shifts right)	P ↓	Q ↑
Supply decreases ↓ (shifts left)	P ↑	Q ↓
Demand increases ↑ (shifts right)	P ↑	Q ↑
Demand decreases ↓ (shifts left)	P ↓	Q ↓
Demand Increases, Supply increases	P ↕ (indeterminate)	Q ↑
Demand Increases, Supply decreases	P ↑	Q ↕ (indeterminate)
Demand decreases, Supply increases	P ↓	Q ↕ (indeterminate)
Demand decreases, Supply decreases	P ↕ (indeterminate)	Q ↓

How Prices Allocate Resources – The Role of Prices

➢ **Prices as Signals in Market Economies**

•In market economies, **prices act as signals** that guide economic decisions.

•They help **allocate scarce resources** efficiently among producers and consumers.

➢ **Balancing Supply and Demand**

•For every good or service, **prices adjust to balance supply and demand**.

•High prices → Encourage **producers to supply more** and **consumers to buy less**.

•Low prices → Encourage **consumers to buy more** and **producers to supply less**.

Example: Cloud Computing Services

•If demand for **cloud storage** increases, providers **raise prices** to manage limited capacity. Higher prices **signal the need for more servers**, encouraging expansion.

•Over time, supply increases, and prices stabilize at a new equilibrium.

How Equilibrium Price Guides Resource Allocation

- **Equilibrium Price Determines Market Outcomes**

The **equilibrium price** plays a crucial role in balancing supply and demand. It determines:
✓ **How much buyers purchase**
✓ **How much sellers produce**
✓ **How resources are allocated efficiently**

- **Efficient Resource Allocation**

  ✓ Ensures **producers make profitable goods**.

  ✓ Encourages **consumers to buy based on needs & budget**.

  ✓ Directs **investment to high-demand sectors**.

Example: Cybersecurity Software Market
Rising demand for cybersecurity software (e.g., due to increased cyber threats, remote work, and cloud adoption) → Prices rise
- Higher prices signal cybersecurity firms (Norton, McAfee, CrowdStrike) to **invest in new threat detection technologies, hire more security experts, and expand their services.**

Supply catches up → Prices stabilize
- As more firms enter the market and existing providers scale up, competition increases, leading to a price **adjustment toward equilibrium**, ensuring businesses can access security solutions affordably.

Outcome:
☑ Prices act as signals to allocate cybersecurity resources efficiently.
☑ Ensures that security software is available without excessive pricing or market shortages.

Chapter Summary

In this chapter, we have

- **Discussed market equilibrium** – how demand and supply interact to determine price and quantity.
- **Discussed market disequilibrium**: Shortage and Surplus
- **Reviewed the impact of shifts in demand and supply** on equilibrium price and quantity.
- **Explored mathematical examples** (linear & non-linear) to determine equilibrium in IT markets.
- **Analyzed changes in equilibrium** when demand or supply increases or decreases.
- **Examined how prices allocate resources** efficiently by signaling production and consumption decisions.

Next…..

- ✓ Elasticity and its Applications